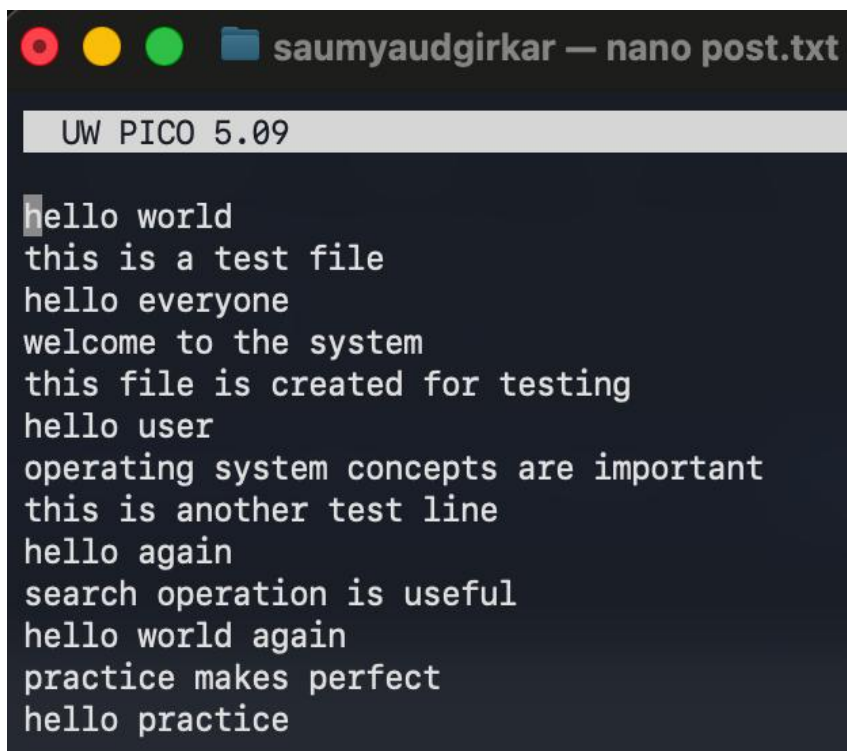


Post-Lab

Aim : To study and implement file handling and text processing in an operating system by developing a C program to search for a specific string in a file and display the results with line numbers.

Content in a file for searching

post.txt

A screenshot of a nano text editor window. The title bar shows 'saumyaudgirkar — nano post.txt'. The editor content shows the following text:

```
UW PICO 5.09

hello world
this is a test file
hello everyone
welcome to the system
this file is created for testing
hello user
operating system concepts are important
this is another test line
hello again
search operation is useful
hello world again
practice makes perfect
hello practice
```

Code :

```
#include <stdio.h>
#include <string.h>

int main() {
    FILE *fp;
    char filename[100], word[50], line[200];
    int line_num = 0, found = 0;

    printf("==== TEXT SEARCH TOOL ====\\n");
```

```

// Input file name
printf("Enter file name: ");
scanf("%s", filename);

// Open file
fp = fopen(filename, "r");

if (fp == NULL) {
    printf("File does not exist!\n");
    return 1;
}

// Input word to search
printf("Enter word to search: ");
scanf("%s", word);

printf("\nSearching for '%s' in %s...\n", word, filename);
printf("-----\n");

// Read file line by line
while (fgets(line, sizeof(line), fp)) {
    line_num++;

    // Check if word exists in line
    if (strstr(line, word) != NULL) {
        printf("Line %d: %s", line_num, line);
        found = 1;
    }
}

if (!found) {
    printf("No match found!\n");
}

fclose(fp);
return 0;
}

```

Output :

```
saumyaudgirkar@Mac ~ % nano search.c
[saumyaudgirkar@Mac ~ % ./search
===== TEXT SEARCH TOOL =====
Enter file name: post.txt
Enter word to search: hello

Searching for 'hello' in post.txt...
-----
Line 1: hello world
Line 3: hello everyone
Line 6: hello user
Line 9: hello again
Line 11: hello world again
Line 13: hello practice
saumyaudgirkar@Mac ~ % █
```